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Celebrity Cruises®

ROYAL CARIBBEAN CRUISES LTD.
EDGE PROJECT

MAS – ELECTRIC POWER PLANT PHILOSOPHY

MAS – PHILOSOPHIE DU SYSTEME DE PRODUCTION D'ENERGIE

STX France

Ouvrage min / *Ship min*	Ouvrage max / *Ship max*	Echelle *Scale*	Section *Section*
J34	Et suivants / *And next*	-	7162
Ce document est applicable à tous les navires compris dans l'intervalle ci-dessus This document supplies to all ship included in the above interval			
Fonction *System*	Reférence historique *Reference number* **M40E4937**	Revision / Effectivité *Revision / Effectivity*	Planche *Sheet*
SNC	Numéro du document du navire de référence *Drawing(s) number(s) of the Reference ship* -	A1	1/34

1. GENERALITIES

1.1. Purpose

The aim of this document is twofold :

- to describe the principles of the electric power plant philosophy for J34 project ;
- to describe the Power Management System (PMS). The PMS is included in the Machinery Automation System (MAS) ;

1.2. Glossary

AFT : Aft
AHU : Air Handling Unit
BO : Black Out
CB : Circuit Breaker
DE : Diesel Engine
DE's : Diesel Engines
DG : Diesel Generator
DG's : Diesel Generators
ECR : Engine Control Room
EDG : Emergency Diesel Generator
ES : Emergency Switchboard
ESD : Emergency Shut Down
FWD : Forward
FZ : Fire Zone
I/O : Input/Output
LO : Lube Oil
MAS : Machinery Automation System
OS : Operator station
PCS : Process Control Station
PEM : Propulsion Electric Motor
PMS : Power Management System
RIO : Remote Input / Output
SFOC : Specific Fuel Oil Consumption
SL : Serial Line
SWBD : Switchboard
UPS : Uninterruptible Power Supply

1.3. Reference documents

Reference documents are the following:

- Electrical distribution diagram : High voltage network ref. 500E2000
- Electrical distribution diagram : Low voltage networks ref. 500E2100
- Electrical distribution diagram : Emergency networks ref. 500E2200
- Electrical production automation ref. 500E1200

2. ELECTRIC POWER PLANT DESCRIPTION

2.1. General

Refer to the following Electrical distribution diagrams :

- Overview electrical system ref. 500E1000
- High voltage network ref. 500E2000 ;
- Low voltage network ref. 500E2100 ;
- Emergency networks ref. 500E2200.

Five main electrical networks are installed on J34 :

1. Main 11 kV-60Hz network : Power supplied by 11 kV generators (5 DG's).
2. Main 440V-60Hz network : Power supplied from 11kV/440V machinery transformers.
3. Emergency 440V-60Hz network: Power supplied by the main 440V-60Hz network or by EDGs (GPP01 and GPP02).
4. 440V-60Hz network for Electric substations and main galleys in FZ6 & FZ7 : Power supplied from 11kV/440V transformers.
5. 400V-60Hz network for Electric substations : Power supplied from same transformers than above for electric substation (11kV/440V/400V transformers).

The power is distributed by means of following six switchboards :

- HMS1 : 11kV Fore High voltage MAIN switchboard Starboard;
- HMS2 : 11kV Aft High voltage MAIN switchboard Portside;
- MS1 : 440V STBD FWD MAIN switchboard ;
- MS2 : 440V PORT MAIN switchboard ;
- ES1 : 440V emergency switchboard 1;
- ES2 : 440V emergency switchboard 2;

2.2. Electric production description

The power generating plant includes the following generators:

- 1 Diesel Generator (11kV ; 60Hz ; 14109 kW ; 600 RPM) and 1 Diesel Generator (11kV ; 60Hz ; 9381 kW ; 600 RPM) connected to the 11kV FORE SWBD (HMS1)
- 1 Diesel Generator (11kV ; 60Hz ; 14109 kW ; 600 RPM) ; 1 Diesel Generator (11kV ; 60Hz ; 9381 kW ; 600 RPM) and 1 Diesel generator (11kV; 60 Hz ; 6543 kW ; 720 RPM) connected to the 11kV Aft SWBD (HMS2)
- 1 Emergency Diesel Generator (440V ; 60Hz ; 1450 kW ; 1800 RPM) connected to the 440V emergency SWBD (ES1)
- 1 Emergency Diesel Generator (440V ; 60Hz ; 1450 kW ; 1800 RPM) connected to the 440V emergency SWBD (ES2)

2.1. 11kV Electric distribution description

The Fore Port main switchboard (HMS1) feeds the following main equipment :

- PTRBT1 Bow thruster 1 starter 3500 kW
- PTRBT3 Bow thruster 3 starter 3500 kW
- HSL1 Ring net reactor 1 7200 kVA
- MPEPHTR1 Port propulsion transformer 1 9200 kVA
- MPEPHTR2 Port propulsion transformer 2 9200 kVA
- VFEGU1B air conditioning compressor 2 1250 kW
- VFEGU1C air conditioning compressor 3 1250 kW
- FHTR1MS1 engine room Port transformer 4500 kVA
- FHTR2MS1 engine room Port transformer 4500 kVA

The AFT STBD main switchboard (HMS2) feeds the following main equipment :

- PTRBT2 Bow thruster 2 starter 3500 kW
- PTRBT4 Bow thruster 4 starter 3500 kW
- HSL2 Ring net reactor 2 7200 kVA
- MPESHTR1 STBD propulsion transformer 1 9200 kVA
- MPESHTR2 STBD propulsion transformer 2 9200 kVA
- VFEGU1A air conditioning compressor 1 1250 kW
- FHTR1MS2 engine room STBD transformer 4500 kVA
- FHTR2MS2 engine room STBD transformer 4500 kVA

3. CONTROL TRANSFER

3.1. General

There are 5 CONTROL STATIONS to control and monitor the power generating plant.

Taking control from each control station implies different control and monitoring capabilities usually defined in automation protocols as "LEVELS", altogether responding to a well-defined "HIERARCHY".

J34 is equipped with a THREE-LEVEL HIERARCHY of control and monitoring and 5 CONTROL STATIONS.

Operator interacts with the PMS, the five diesel engines by means of this THREE-LEVEL HIERARCHY from one of the following CONTROL STATIONS :

FWD Diesel engine room	LEVEL1
Aft Diesel engine room	LEVEL1
FWD main 11 kV switchboard	LEVEL2
AFT main 11 kV switchboard	LEVEL2
ECR	LEVEL3

◊ The principles of control transfer are described hereafter for the four DGs.

◊ Each generator is <u>independently</u> controlled.

◊ It is possible to have one generator controlled from one level and another generator controlled from a different level.

The control levels for the PMS are considered in order of priority:

- **Level1 (highest priority) :**

  Local control panel DG1 located at the engine (To manually control DG1);
  Local control panel DG2 located at the engine (To manually control DG2);
  Local control panel DG3 located at the engine (To manually control DG3);
  Local control panel DG4 located at the engine (To manually control DG4);
  Local control panel DG5 located at the engine (To manually control DG5);

- **Level2 (intermediate priority) :**

  FORE SWBD HMS1 mimic panel located in the FORE HV switchboard room (To control DG1 and DG2);

  AFT SWBD HMS2 mimic panel located in the AFT HV switchboard room (To control DG3, DG4 and DG5);

- **Level3 (lowest priority) :**

  MAS operator stations located in ECR

  Note : MAS operator stations located in Wheelhouse

  In normal condition : Access control will be limited to certain function as Ballast and fire system mimics

  In Emergency situation : Full access control through a password

  Refer to "MAS – Information treatment" M40E4801

<u>THE CHANGE OF CONTROL LEVELS COMPLIES WITH THE FOLLOWING RULES:</u>

- Only one control level is active at any time for each genset.
- Changing level has no effect on actuators status.

- Changing level has no effect on monitoring; it is always available at any level.
- Changing level may have effect on PMS automatic sequence capabilities and automatic load sharing.

3.2. Principle of control transfer for diesels generators

All control transfers are possible. Thus:

- Level1 → Level2
- Level1 → Level3
- Level2 → Level1
- Level2 → Level3
- Level3 → Level2
- Level3 → Level1

A drawing which shows the speed control hierarchy and the start/stop hierarchy is attached in Annex 1.

3.2.1. Level 1 (DG room)

Each DG has its DE Local Control Panel which is equipped with a hardwired Local/Remote switch :

If the operator switches the Remote/Local switch to "Local", the control of the corresponding DG is taken away from any other level, without any transfer sequence excepted alarms which are always managed by MAS.

Remote alarms and monitoring capabilities remain available in MAS.

The following main actions are controlled from the Local Control Panels of DE1 to DE5 :

- Diesel Engine Start/stop;

If the operator switches the Remote/Local switch to "Remote", control is transferred to Level 2 or Level 3 depending on the control selector "Local/Remote" position located in corresponding SWBD.

3.2.2. Level 2 (HV switchboard mimic panel)

Control at the switchboard room mimic panel is available if the local Level 1 switch of one or more DE local switches located on each DE local control panel is in "Remote" position.

Thus control of DGs at switchboard room is possible and can be taken by switching to "Local" the Local/Remote selector (**<u>One control selector for all DGs connected to the SWBD</u>**) located on the HV SWBD mimic panel.

The following actions are available on the FWD and Aft SWBD room mimic panels :

- Start/Stop of each DG
- Increasing/decreasing of each DG speed
- Circuit breaker selection
- Manual or Auto synchro
- Opening/closing of each DG circuit breaker

To transfer to Level 3, operator has to switch to "Remote" the Local/remote selector located on the HV SWBD mimic panel.

To transfer to level1, operator has to switch in "local" the local/remote switch located on DE local control panel.

3.2.3. Level 3 (MAS)

Control through MAS, for each individual DG, and for PMS functions is available if :

- Local/remote control selector located on the HV SWBD mimic panel is in "Remote";
- Local/remote control switch located on the DE local control panel is in "Remote" position.
- MAS is functioning without major malfunctions.

Switching transfer activation from Level3 to Level2 or to Level1.

- Level 3 → Level 2 : Level 2 "takes control of all DGs by switching to "Local" the Local/Remote selector on the HV SWBD mimic panel (switch in level 1 has to be still in remote). The transfer operates immediately for the corresponding DGs.
- Level 3 → Level 1 : Level 1 "takes control" by switching to "Local" the Local/Remote switch on DE local panel. The transfer operates immediately for the corresponding DG.

For the available actions through the MAS, refer to chapter 4.

3.2.4. MAS failure

In case of MAS failure (refer to MAS documentation), an alarm is activated and control capability and manual transfer controls are effected as follows :

- If control is in Level 1: Control stays in Level 1 and it can only be transferred to Level 2. Level 3 cannot take control.
- If control is in Level 2: Control stays in Level 2 and it can only be transferred to Level 1. Level 3 cannot take control.
- If control is in Level 3: Control must be transferred manually to Level 2 or to Level 1.

3.3. Principle of control transfer for 11kV switchboards

Each AFT and FWD SWBD is equipped with a Local/remote switch.

This Local/remote switch is only used for all SWBD incoming, that means DGs and HV cross tie.

SWBD outgoing which are not belong to the Local/Remote switch are : Machinery transformers, propulsion transformers, sub/stations ring net, thrusters and AC compressors.

Whatever the 11 kV cross ties position, the total set of generators and HV cross tie of one HV switchboard can be transferred in SWBD Local control while the other remains in fully automatic control by MAS.

When 11 kV cross ties are opened, one switchboard can be in Local control while the other remains in fully automatic control by MAS.

When the 11 kV cross ties are closed, the manual transfer to local control of one switchboard will allow the other switchboard to remain controlled by MAS.

4. POWER MANAGEMENT SYSTEM

The PMS is an integral part of MAS and will consist of the following main functions.

1. Manual or Automatic START/STOP of the main generators (DG)
2. Control of the electrical power supply and distribution of the Ship's electrical network consisting of two main switchboards HMS1 and HMS2.
3. Provision of the necessary energy for the electrical propulsion system.
4. 'Switching off' of non-essential consumers in case of overload in the Ship's electrical network according to the actual selected configuration of the main switchboard (Load shedding of chillers, air conditioning equipments and Galleys).
5. Load demand system for heavy consumers in the individual parts of the Ship's electrical network according to the actual selected configuration of the main SWBD for :
   - Thrusters
   - Chillers
6. Control of the 11 kV consumers in the main SWBD.
7. Automatic restoration of the Ship's electrical network after a blackout.
8. Monitoring of HMS1 and HMS2 SWBD and display of main values and alarms for Main Low voltage switchboards.

4.1. Diesel generators (DG)

Safety functions of Diesel Engines are achieved in the Diesel control system, which takes care of the shutdowns and start blocking conditions(*) of the Diesel. Each Diesel has its own control system.

(*): Start blocking conditions due to generator/s conditions are achieved by MAS.

All essential signals needed to operate the Diesel (Start/Stop command, summary shutdown status …) are hardwired.

A serial line from each Diesel control system is connected to MAS to send relevant measurements and status.

For each DE the following functions achieved by MAS, are provided:

- Manual Start/Stop from keyboard;
- Automatic Start/Stop;
- Standby selection;

4.1.1. Diesel generators : Manual Start / Stop

The operator can start manually each DE from the keyboard if there is no start blocking condition.

Operator can also stop manually from the keyboard. This command (achieved by PMS) will:

- Unload the generator
- Open circuit breaker when generator load is less than 5%. Diesel will run without load for X minutes (cooling time adjustable at keyboard) before stopping.

It is possible to interrupt a stopping sequence by a keyboard action.

4.1.2. Diesel generators : Automatic Start / Stop

The automatic starting of standby DG can be activated by the following:

- Standby start
- Load dependent start
- Blackout start sequence

Automatic starting is limited to DE which are "ready to start" and selected as "Auto".

Standby start: When autostop (*) or shutdown condition occurs, MAS will start and connect the next standby generator according to the standby sequence selected by operator at keyboard.

(*): Autostop corresponds to the "load reduction requested" conditions established by the Diesel control system.

The faulty DG will be disconnected and then stopped X seconds after the standby DG started has reached "Connected" condition.

Load dependent start: If the actual load is higher than a pre-set limit over a certain time duration, MAS will start and connect the next standby generator. Pre-set values are settable from keyboard.

In addition a load dependent start is activated if the DG current exceeds 95% of nominal current fore more than 5 seconds. The parameters setting mentioned above are adjustable by operator.

Start by blackout sequence: In case of blackout, MAS will start and connect a certain number of generators; refer to relative §.

The automatic stopping of running DG can be activated by the following:

- Stop of faulty Diesel
- Load dependent stop (in accordance with operating mode)
- Shutdown

Automatic stopping is limited to DE which are selected as "Auto".

Stop of faulty Diesel: When autostop (refer to above) occurs, MAS will first start the next standby generator and then stop the faulty Diesel engine after load transfer (X seconds after RUN condition).

Load dependent stop (in accordance with operating mode): If the actual load is lower than a pre-set limit over a certain time duration, then MAS will stop and disconnect the next generator to be stopped.

Shutdown : Will stop immediately the DE, MAS will automatically start and connect next standby generator.

4.1.3. Diesel generators Auto / Manu selection

The "Auto / Manu" selection is possible from MAS keyboard for each Diesel generator.

Selection "Auto" is required for automatic starting and stopping of a generator by the PMS functions.

4.1.4. Diesel engines slow turning

The slow turning sequence is integrated in the start sequence managed by the Diesel local control system.

The slow turning is initiated only when the diesel is in remote control or in Local mode.

Cyclic /periodical slow turning for DG1 to DG4 : An automatic slow turning is initiated every 30 minutes when signal "Remote standby request" is activated from MAS and sent to diesel control system. This signal is activated in MAS when Diesel is selected as standby in MAS (no start blocking and selected in Auto) and Diesel is the next standby to be started.

Slow turning before starting for DG1 to DG5 : An automatic slow turning (two revolutions within 10 seconds) is initiated before each starting of diesel engine if diesel engine is stopped since more than 30 minutes, except when diesel is started by blackout restarting sequence, the Diesel control system will do a direct starting without slow turning.

In case of slow turning failure, an alarm is activated (signal coming from serial line) and a shutdown condition is initiated.

4.2. PMS functions achieved by MAS

The main automation functions achieved by MAS for the two main 11 kV switchboards are the following:

- Automatic synchronising (#)
- Control of 11 kV bus tie circuit breakers
- Load sharing and Frequency control (*)
- Generator priority selection
- Load increase control
- Load dependent Start/Stop (#)
- Blocking of heavy consumers (*)
- Load shedding (preferential trip) (*)
- Mode selection
- Automatic restarting after blackout (*)

(*) : Global automatic function which can be selected "on" or "off".

(#) : Automatic function which can be selected "on" or "off" for each diesel generator by the Diesel Auto/Manu selector

4.2.1. DGs automatic synchronising

The automatic synchronising of generators is performed by using fully automatic synchro-unit located in the main switchboard.

MAS selects which generators have to be synchronised to the bus bar by giving a synchronising command to the synchro-unit.

If synchronising exceeds 90 seconds, a synchronising time out alarm is activated. If the initial synchronising command came from an automatic starting command, another generator will be started.

During normal operation, the synchronising command is active when:

- Synchronising is selected to automatic mode
- The generator is running in MAS control mode
- The generator is not connected to the bus bar
- No other generator is already in synchronising sequence
- Generator breaker not in alarm condition
- Generator and breaker in remote control
- Main switchboard in remote control
- HV bus tie circuit breaker (from the same switchboard) not tripped

During emergency operation (restart after blackout for example), the first generator is connected directly without any synchronising.

4.2.2. Control of 11 kV bus ties circuit breakers

The control of the bus ties circuit breakers through the MAS is manual by the operator.

Both bus tie circuit breakers located in HMS1 and in HMS2 have synchronising function.

When a MAS closing command, the automatic synchronising of HMS1 and HMS2 main switchboard is performed by use of full automatic synchro-unit located in the main switchboards (one synchro unit per SWBD). Synchronisation is bypassed when connecting to "dead busbar" (local automation function).

If synchronising exceed 90 seconds, a synchronising time out alarm is set.

Opening command of the bus tie circuit breaker by the operator will be executed with load sharing and current limitation to be done first. This function will lead an automatic starting of generator if the current in the bus tie cannot decrease under a pre-set value (about 3% of the nominal current in the bus tie).

The MAS send the open order to the circuit breaker when the bus tie load is less than a pre-set value.

Note : If minimum load is not reached after a time delay, an alarm will be raised and the opening command will not be completed.

Only one command (open/close) can be executed from MAS at a time for the both HV circuit breakers (QCTHM1, QCTHM2).

4.2.3. DGs load sharing / Priority selection & Frequency control

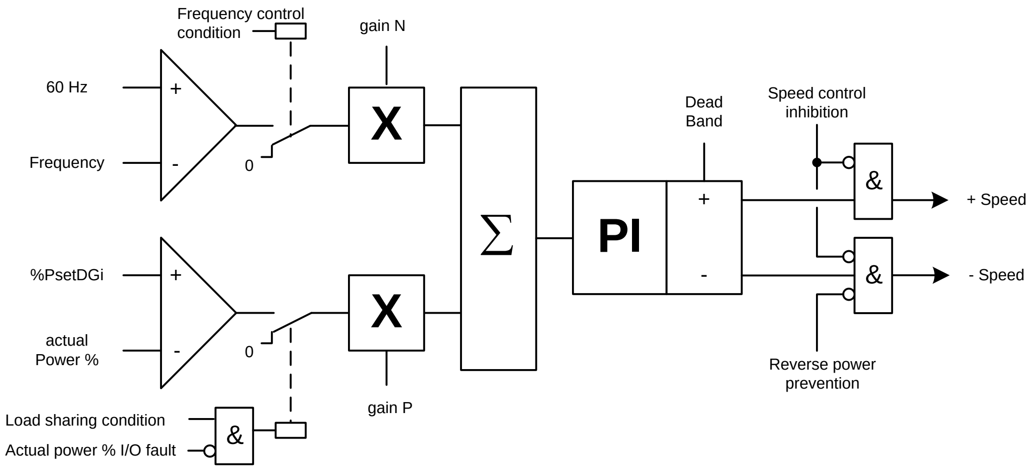
4.2.3.1. DGs load sharing

Load sharing will be based on speed droop mode for all DG independent of the control mode selected.

Automatic load sharing and frequency control function from MAS are active if MAS is in control.

Based on the net frequency and generators load, MAS controls DE speed set points.

Net frequency is compared with the frequency set point 60 Hz and the generator actual load (Current) is compared with calculated load set points: %PsetDG(Current).



MAS will perform the load sharing by sending up and down pulses to the Diesel control system for DG.

Unbalanced mode :

When the HV bus ties are closed and DG with priority 1 is connected then it is possible to select the unbalanced mode.

The unbalanced mode consists to fix the power set point of the master DG (DG with priority 1) when the other DGs connected (called slave DGs) take the remaining load in equal sharing . Power set point of the master DG can be adjusted from 20% to 100%.

When slave DGs power set point goes lower than 10% or reach 100%, then equal load sharing is applied to all DGs

When HV bus ties are open, unbalanced mode can be selected on each group of diesel generators with the same philosophy as described above (both group of DGs FWD and Aft can be in unbalanced mode or only one group of DGs or both can be in equal load sharing).

Sea Optimal mode :

In this Sea Optimal Mode, load sharing set point for each engine connected will be calculated taken into account the optimized engines SFOC. It means that each individual engine can have different load set point based on the most optimal combination of load set points.

4.2.3.2. HV switchboards frequency control

Based on bus bar frequency and generator load, MAS controls the generator speed by sending up and down pulses to each generator speed control unit. The bus bar frequency is compared with the frequency set point 60 Hz. A dead band is provided (+/- 0.3 Hz) to reduce continuous hunting on governor.

The speed control unit of generator controls engine speed.

When bus tie is open, MAS will activate frequency control for each bus bar independently.

4.2.3.3. Diesel generator priority selection

The operator can define the priority (1 to 5) of the generators, priority 1 being the highest priority. This priority will be used in the PMS function for the automatic start and stop sequences and in the restart sequence after blackout.

4.2.4. DGs load increase control

After coupling of a generator , MAS will softly load the generator according to the load ramp rate (approximately 6 minutes to get full load) set in the MAS, except in emergency situations which are after blackout and when a standby start due to a shutdown or a faulty engine is activated.

4.2.5. DGs load limitation

For each DG, operator can set two different load limitations.

The first is a real load limitation, achieved by setting a power limit in MW through a PMS graphic page. In this case the corresponding generator load is shared up to the preset power limitation value.

The second load limitation corresponds to a *de-rating condition* of the generator's nominal power.

The de-rating condition is achieved by setting "Power rated" in % through a PMS graphic page. In this case the corresponding generator will be loaded in the ratio "Power rated/Nominal power".

Note : Power limitation of each generator [minimum value of both above limitations] is sent to the propulsion control system.

4.2.6. DGs load reduction

Any "load reduction request" condition coming from diesel local control system will be considered and executed in the MAS as following :

Either as a "delayed shutdown" allowing automatic starting and connecting to network of the next stand-by DG (if any Standby DG available) before stopping the faulty DG.

Or as a real load reduction of the faulty DG by using the MAS de-rating power function (setting to 50% of nominal power) if there is not any Standby DG available to be started.

Note : Load reduction power setting by generator derating power can be adjusted by the operator from 40% to 70% of nominal power (default setting : 40%). Also, time delay between load reduction request and PMS load reduction action can be adjustable by the operator from 0 to 300 sec (default setting : 4 sec).

4.2.7. DGs load dependent Start / Stop

MAS calculates the bus bar's Nominal Power, measures the Actual Load for each generator, and then calculates the Power Available.

Available Power = (Bus bar's Nominal Power) minus (Total Load on bus bar) minus (Reserve Power)

Reserve Power = 0 when in 'Open Sea' mode or 'Harbour' mode,

or equal to (Bow thrusters Nominal Power) minus (Bow thrusters Actual Power) when in 'Manoeuvring' mode or thruster connected or thruster start request.

The Available Power is compared with the load dependent start and stop limits.

If the load increases so that the actual load exceeds 85% (start limit adjustable), the available power is less than 15%. After 30 seconds (adjustable) in this condition, the standby generator will start and automatically synchronise to the bus bar.

In addition a load dependent start is activated if the DG current exceeds 95% (start limit adjustable) of nominal current fore more than 5 seconds (adjustable).

If the load decreases so that remaining synchronised generators load (means load after stopping a DG) is less than 75% (stop limit adjustable) for a period of 5 minutes (adjustable), then the last generator connected to the net (according to starting sequence selected) will automatically be downloaded and disconnected.

If bus tie is open, MAS will calculate Available Power on each side and load dependent Start/Stop function will be activated on each bus bar (AFT and FWD switchboards).

The load dependent stop function is blocked in "Manoeuvring" mode.

In "Open Sea" mode, auto stop function is blocked when two generators remain connected.

4.2.8. Blocking of heavy consumers

Blocking of AC compressors :

Three AC compressors (1250 kW each) will be taken into account by the blocking of heavy consumers function achieved in the MAS. If there is not enough power available (after receiving a start request) to start one of these motors, the MAS automatically start up and connect the next standby generator.

Note : One large generator (DG1 or DG3) is needed on bus bar to start an AC compressor. If there is not any large generator connected then two (2) generators (large, medium or small) have to be connected.

Blocking of Bow thrusters :

Four bow thrusters (3500kW each) will be taken into account by the blocking of heavy consumers.

When a thruster start is requested, the MAS checks the available power on the bus network ; the start blocking signal is activated if there is no enough power available (Pavailable < Pstartlimit) on the net or if not enough generator connected (two generators (*) must be coupled for starting a thruster and generators load must be equal with +/- 15%).

(*) : One large generator (DG1 or DG3) and one medium generator (DG2 or DG4) have to be connected to allow thruster starting

Moreover when manoeuvre mode is selected, the MAS makes power reservation for those four thrusters, which is for each individual thruster :

When thruster CB connected then power reservation is 3,5 MW minus the bow thrusters measured consumed power

This to ensure that sufficient generators are on the net when the Ship is in manoeuvring.

Note : Blocking heavy consumers function will guaranty that if there is several start demands at the same time for different heavy consumers then only one start authorisation will be active at a time (time delay of few seconds between each start authorisation).

Note : When blocking of heavy consumers function is OFF (PMS selector "Blocking Heavy" in manual) then thruster start interlock is still valid if there is only one generator connected (equal load +/- 15% will be not taken into account).

4.2.9. Load shedding (preferential trip)

Load shedding performed by MAS is automatically activated in harbour, manoeuvre and open sea mode in case of:

- Generator overload
- Low frequency on bus bar

Preferential tripping function is activated in two stages:

1. Cut off power to galley and AC compressors (1)
2. Cut off power to Accommodation ventilation (2)

(1) : Preferential trip 1 : MAS order sent by hardwired output to HV SWBDs for AC compressors and to GSUB6, GSUB7, PSUB2, PSUB4 and LSUB7 for main galleys.

(2) : Preferential trip 2 : MAS order sent to ESD through system networks for accommodation ventilation.

Note : MAS load shedding function will take into account the HV bus tie circuit breaker position. If the HV bus ties are opened then one load shedding function will operate for each HV SWBD.
Load shedding function can be inhibited by the operator with a selector switch located on an MAS mimic.
Alarm will be raised "Preferential tripping 1" and "Preferential tripping 2" corresponding to the two different stages of the load shedding, this for each HV switchboard (case of HV switchboards uncoupled).

MAS preferential tripping actions will require a reset action by the operator through a PMS mimic. This reset action will be allowed only when normal situation is recovered (hysteresis).

4.2.10. Power plant mode selection

The power plant mode is selected by operator according to Ship's operation.

The operation mode selection is achieved by means of a software 3-position switch located on MAS operator station.

"Harbour/Manoeuvring/Sea/Optimal Sea".

Each position corresponds to a pre-established minimum configuration of the power plant. This configuration is automatically performed by MAS. This means that :

- The load dependent stop function will be blocked when the minimum configuration is reached.
- When mode transfer is operated, the automatic connection of the next standby generator is obtained if the minimum configuration is not reached.

Harbour mode:

One DG is sufficient (whatever summer/winter seasons and thus depending to load analysis) but the operator may connect more generators depending on the network configuration. However the PMS function will automatically connect the next standby generator if required.

Manoeuvring mode:

With the bus ties open : 2 DGs connected on each HV SWBD (one large and one medium generator)

With the bus ties closed : 2 DGs connected (large or medium generators),

Note : All load dependent stop functions are blocked in this mode.

The operator can decide to operate the power plant with the bus tie open in manoeuvring mode.

Sea mode :

With the bus tie open : 2 DGs connected on each HV SWBD (one large and one medium generator)

With the bus tie closed : 2 DGs connected (large or medium generators), Note : Load dependent stop functions are blocked when the **minimum configuration "Two generators connected"** is achieved whatever bus ties configuration.

Optimal Sea mode :

Optimal Sea mode will have same DGs configuration as in Sea mode but load sharing will be different (refer to § 4.2.3.1).

4.2.11. Thrusters pitch limitation

In order to prevent a black out due to the big amount of thrusters power, thruster maximum pitch will be limited and thus generator load in case of the minimum configuration of the power plant is not reached, which correspond to :

- When HV cross ties are closed : Thruster max pitch limitation if number of generators connected < 2 DGs (whatever the size Small, Medium or Large of the DG connected) or if the two (2) DGs connected are a Medium engine and the Small engine.
- When HV cross ties are opened : Thruster max pitch limitation if number of generators connected < 2 DGs (whatever the size Small, Medium or Large of the DG connected) or if the two (2) DGs connected on HMS2 are a Medium engine and the Small engine.

An individual pitch limitation order (binary signal) will be sent by MAS to each thruster control panel. Thruster pitch limitation will be executed, according to the MAS demand, by the thruster control system as a de-rating of the maximum pitch.

4.2.12. Propulsion circuit breaker control

As for AC compressors and thrusters, the 11kV propulsion circuit breakers will not be controlled from MAS. Each 11kV circuit breaker to power supply propulsion transformer will be controlled from the propulsion control system.

There is not any Propulsion circuit breakers "Authorization to close" sent by MAS to the propulsion control system because of the pre-energizing of the propulsion transformers.

Note : For the automatic restarting after blackout, the propulsion transformer circuit breakers will be also controlled by the propulsion control system using the same philosophy (no "closing authorization" from PMS).

Propulsion control system will reclose propulsion circuit breaker after black out when :

- Two (2) generators are connected (among DG1 to DG5) on the bus bar of the corresponding propulsion circuit breaker (interlock done in propulsion control system)
- The operator put the propulsion levers references at 0 rpm
- The auxiliaries restart automatically by propulsion control cabinet 30 seconds after first generator connection in the HMS
- The operator reset the propulsion system in control panel

4.2.13. Electrical substation loop configuration

The electrical substation loop configuration can be done through the MAS.

All circuit breakers which are with remote control, means HMS1/HMS2 outgoing and substation incoming, will be manually control by the operator through MAS.

Each remote controlled circuit breaker has its own local/remote switch.

Nominal configuration of the loop is as following :

- Electrical substation supplied by HMS1 (Q31HM1 closed) and by HMS2 (Q32HM2 closed)
- Open loop configured : one of the following CBs is opened QS1RM2, QS23RM3, QS34RM4, QS45RM5, QS56RM6 and QS67RM7.

To change the open loop configuration, the operator will have to do the following :

- If HV bus ties HMS1/HMS2 are closed :
  - Operator manual closing of the last circuit breaker open of the loop, that will lead closed loop configuration
  - Warning message on the HV loop mimic "HV loop has to be opened"
  - Operator manual opening of one circuit breaker of the loop in order to achieve the new loop configuration expected
  - An alarm will be raised in MAS "HV loop to be configured open" if close loop detected since 30 sec.

    Note : A safety protection located in protection relay of HV outgoing circuit breakers (inside one of both circuit breakers for a loop) will open the corresponding circuit breaker when close loop is detected since 90 sec.
- If HV bus ties HMS1/HMS2 are not closed :
  - Closing of the last circuit breaker of the HV loop is interlocked (MAS internal logic).
  - Operator manual opening of the circuit breaker which correspond to the new HV loop configuration, that will lead blackout of a part of the electrical substations.
  - Operator manual closing of the circuit breaker(s) to reach the new HV loop configuration.

4.2.14. Distribution circuit breaker control

In addition of some HV outgoing circuit breakers already described in the present document, circuit breakers which are controlled from MAS are identified in diagram in annex 2.

5. BLACK OUT

Two different situations are taken into account for power restoration after blackout:

- Total blackout
- Half blackout.

Main electrical network is shown in Annex 2.

5.1. Total black out

Initial situation (Normal operating condition):

- Bus tie breakers HMS1/HMS2 are CLOSED. Switchboards are fed by diesel generator(s).
- MS1 is supplied by 11kV/440 VAC transformers FHTR1MS1 and FHTR2MS1.
- MS2 is supplied by 11kV/440 VAC transformers FHTR1MS2 and FHTR2MS2.
- Internal bus tie between in MS1 and in MS2 are OPEN.
- Emergency switchboard ES1 (440V) is fed by MS1.
- Emergency switchboard ES2 (440V) is fed by MS2.
- Bus tie between ES1 and ES2 is OPEN.

Blackout situation:

1. All Diesel generator circuit breakers are OPEN and bus bars HMS1 and HMS2 are DEAD.
2. Bus tie between HMS1/HMS2 is CLOSED.
3. Automatic opening by "Voltage loss" of all 11kV outgoing circuit breakers.
4. Voltage loss in MS1 and MS2 switchboards.
5. Opening (after time delay) of incoming circuit breaker in ES1 (from MS1/ES1)
6. Opening (after time delay) of incoming circuit breaker in ES2 (from MS2/ES2)
7. Detection of voltage lost in ES1 and ES2 (relay on bus bar).
8. Automatic opening by "Voltage loss" of ES1/ES2 bus ties (if closed prior to blackout)

Automatic power restoration sequence:

This automatic power restoration sequence is as follows:

1. Voltage restoration in ES1 and in ES2 is done by local automation.
2. Voltage restoration in HMS1 and/or HMS2; then in main low voltage switchboards MS1 and MS2. This is done by MAS.
3. Reconfiguration of the electrical distribution to as near as possible what it was prior to the Blackout.

   This power restoration after Blackout is performed by MAS and is divided in two parts:

   - Restarting of the main electrical production plant
   - Restoration of the electrical distribution plant.
4. Reconfiguration of ES1 and ES2 power supply when automatic restart after Blackout sequence is completed is performed **BY OPERATOR**.

Conditions for automatic restarting by MAS of the HV power plant are the following :

- "Blackout recovery" selected in ON (software switch located on MAS PMS overview mimic) (*)
- On HMS1 : All generator CBs open and bus bar dead
- On HMS2 : All generator CBs open and bus bar dead

(*) : The alarm "Blackout recovery locked "will be activated if "Blackout recovery" is selected in OFF.

5.1.1. ES voltage restoration after Blackout

When ES1 "Dead bus bar" status is detected and all incoming circuit breakers of ES1 are open, the EDG1 is automatically started and connected to the bus bar (done by local automation). Same as above for ES2 and EDG2.

Impaired situation :

EG1 start failure during voltage restoration sequence:

During the voltage restoration sequence, if there is a start failure of the EG1 or a closing failure of the EG1 CB or if EG1 is blocked, then the following will be realised by MAS :

1. Trip commands for non-Solas consumers sent to both emergency switchboards ES1 and ES2 (this 30 sec after blackout detection on ES1 bus bar, only 3 sec if EG1 blocked).
2. Closing of the ES1/ES2 bus ties 45 sec after blackout detection on ES1 bus bar (only 5 sec if EG1 blocked). Both emergency switchboards are now supplied by EG2.

Note : The above MAS automation function will be also realised when only ES1 is on blackout situation (ES2 still alive)

Closing of ES1/ES2 bus ties is blocked if ES1 short circuit. ES1 short circuit corresponds to at least one of the three incoming CBs tripped (QS1ES1, QS2ES1 and QCTES1).

EG2 start failure during voltage restoration sequence:

During the voltage restoration sequence, if there is a start failure of the EG2 or a closing failure of the EG2 CB or if EG2 is blocked, then the following will be realised by MAS :

1. Trip commands for non-Solas consumers sent to both emergency switchboards ES1 and ES2 (this 30 sec after blackout detection on ES2 bus bar, only 3 sec if EG2 blocked).
2. Closing of the ES1/ES2 bus ties 45 sec after blackout detection on ES2 bus bar (only 5 sec if EG2 blocked). Both emergency switchboards are now supplied by EG1.

Note : The above MAS automation function will be also realised when only ES2 is on blackout situation (ES1 still alive)

Closing of ES1/ES2 bus ties is blocked if ES2 short circuit. ES2 short circuit corresponds to at least one of the three incoming CBs tripped (QS1ES2, QS2ES2 and QCTES2).

EG1 and EG2 start failures during voltage restoration sequence:

During the voltage restoration sequence, if there is a start failure of the EG1 or a closing failure of the EG1 CB and similar problem on EG2, then operator may go locally to clear the failures cause.

5.1.2. Restarting of the electrical production plant

5.1.2.1. General

The restarting of the electrical production plant is done by MAS and consists of starting all available DGs from DG1 to DG5 (whatever the mode "Harbour/Manoeuvring/sea") and connecting of a certain number of generators on the 11 kV network.

The number of generators to be automatically connected will depend on the selected mode "Harbour/Manoeuvring/Open Sea".

The choice of DG and the starting order depends on:

1. Availability of the bus bars (no major faults, such as a short circuit)
2. Number of generators to be connected;
3. Availability of generators (The selection "Manual" of the DG by the operator will lead start blocking of the corresponding DG even for the automatic restarting sequence after blackout);
4. Diesel FO configuration OK (either in HFO or in MGO)
5. Priority selection of generators

General rules applicable to all sequences:

1. All generators (depending on above conditions) will start up at the same time (5 sec delay between each DG starting order);
2. The first generator to reach nominal voltage will be connected on dead bus bar without synchronising. (MAS interlock to be provided to ensure that only one generator can be connected on dead bus bar);
3. Generator already running will be used in the power restoration sequence if its circuit breaker is not tripped;
4. After connection of the first generator, the other(s) running alternators will be synchronised (one at a time);
5. In case of start failure or synchronising failure, another generator will be connected;

<u>Design of DG fuel oil feeding systems for a BO situation :</u>

During blackout, all diesels HFO/MGO feeder and booster pumps are stopped.

The FO pneumatic pump CMM47A (CMM47B) will be started by the MAS by opening the air valve KAA2217 (KAA2215).

When the FO pneumatic pump is running, DGs (one or two depending of Harbour/Manoeuvre/Sea mode selection) will be automatically started according priority selection described above.

The FO pneumatic pump will be automatically stopped by the MAS by closing the valves on air. This stop order will be executed when the boost pump of DGs restarted after blackout has been checked in running status.

The feeder pumps (one of each couple of pumps on FWD and aft feed unit module) will be automatically restarted after black out if one of each couple of pump was running prior to the black out.

The boost pumps will be automatically started if the corresponding diesel has been restarted after black out.

5.1.2.2. In harbour mode

Note : All available generators will be automatically started when blackout is detected.

In harbour mode, one generator is sufficient to take the total load, so only one DG will be automatically connected (according to DG priority) when HV bus ties are closed. If HV bus tie HMS1/HMS2 is open, then one DG will be automatically connected on HMS1 and another DG will be automatically connected on HMS2.

Blackout condition to be read by MAS : **(Time: T0)**

- BO recovering function selected in ON in MAS (*).
- PMS mode selector switch in "Harbour" position.
- All generator circuit breakers OPEN and dead bus bar in HMS1 and HMS2.
- Pneumatic pumps CMM47A and CMM47B started

(*): A critical alarm will be activated when BO recovering function is not ON.

MAS generator restarting sequence :

- Starting of the all available DGs.
- Closing of the first standby DG circuit breaker.
- Remaining restart sequence after BO to restore electrical distribution (refer to chapter 5-1-3-2).

5.1.2.3. In manoeuvring mode

Note : All available generators will be automatically started when blackout is detected.

Two (2) generators will be automatically connected with at least one big generator (DG1 or DG3, according to DG priority) when HV bus ties are closed. If HV bus ties HMS1/HMS2 is open, then two (2) DGs will be automatically connected on each HV switchboard HMS1 and HMS2, those DGs being DG1, DG2, DG3 and DG4.

Blackout condition to be read by MAS : **(Time: T0)**

- Blackout recovering function selected to ON in MAS (*).
- Mode selector switch is NOT in "Harbour" position.
- All generator circuit breakers OPEN and DEAD BUS BAR condition in HMS1 and HMS2.
- Pneumatic pump CMM47A and CMM47B started

(*): A critical alarm will be activated when BO recovering function is not ON.

MAS generator restarting sequence :

- Starting of all generators (1).
- Connection of 2 (two) generators with at least one big generator (if HV bus tie closed). Same for each HV switchboard when HV bus tie are opened. Choice of generators to be connected according priority selection and availability of generators.
- Closing the circuit breaker on dead bus bar of the first generator reaching the nominal voltage.
- Synchronising of the other generators.
- Continuing the restart sequence after BO to restore electrical distribution (refer to chapter 5-1-3-3).

(1) : Start order will be sent to each available generator with 5 sec delay between each start order

5.1.2.4. In sea mode or optimal sea mode

Same as in manoeuvring mode.

5.1.3. Restoration of the electrical distribution plant

5.1.3.1. General

The restoration of the electrical distribution plant during BO restart sequence is done by MAS and consists of:

- re-closing most of 11 kV feeder circuit breakers (excluding AC compressors, thrusters and propulsion).
- supplying power to all main 440V switchboards to restart essential fans and pumps.

5.1.3.2. In harbour mode

The restoration of the electrical distribution plant in harbour mode will be achieved in the following order:

Initial condition is one generator connected on 11 kV bus bar (Time: T1) (refer to § 5.1.2.2)

<u>1st Step : supply of 440V switchboards by MAS</u>

- **supply MS1-1 – Fore Portside main switchboard**
  - Closing order Q11HM1 to power supply FHTR1MS1 (T1+X1)
  - When Q11HM1 closed, closing order QS1MS1 to power supply MS1-1 (T1+X2)
- **supply MS1-2 – Fore Portside main switchboard**
  - Closing order Q12HM1 to power supply FHTR2MS1 (T1+X3)
  - When Q12HM1 closed, closing order QS2MS1 to power supply MS1-2 (T1+X4)
- **supply MS2-1 – Aft Starboard main switchboard**
  - Closing order Q11HM2 to power supply FHTR1MS2 (T1+X5)
  - When Q11HM2 closed, closing order QS1MS2 to power supply MS2-1 (T1+X6)
- **supply MS2-2 – Aft Starboard main switchboard**
  - Closing order Q12HM2 to power supply FHTR2MS2 (T1+X7)
  - When Q12HM2 closed, closing order QS2MS2 to power supply MS2-2 (T1+X8)

With value for X1 = 2 sec ; X2 = 4 sec ; X3 = 6 sec until X8 = 16 sec.

Note1 : The secondary transformer circuit breaker will be automatically closed as mentioned above only if all incoming CBs of the LV bus bar (to be applied for each LV bus bar : MS1-1; MS1-2, MS2-1 and MS2-2) are not tripped.

Note 2 : The essential auxiliaries (fans and pumps) will be restarted according to the sequence described in "Electrical production and distribution ref. M40E4929". Auxiliaries to be restarted after BO are grouped according to power supply source (emergency source, normal source, with/without changeover of the power supply). Each group has its own restarting sequence.

<u>2nd Step : supply of electrical sub-stations loops by MAS</u>

- **supply Loop sub-station 2 to 7**
  - Opening order on all incoming CBs (*)of the HV loop (T1+X9)
  - Closing order Q31HM1, CB which supply HV loop from HMS1 (T1+X10)
  - Closing order Q32HM2, CB which supply HV loop from HMS2 (T1+X11)
  - Sequential closing order of all HV loop incoming CBs (*) (T1 + X14 => X19)

    (*) : QS1RM2, QS23RM3, QS34RM4, QS45RM5, QS56RM6 and QS67RM7

    Note : X9 = 18 sec ; X10 = 20 sec ; X11 = 22 sec ; X14 to X17 : X14 = 28 sec ; until X17 = 34 sec

<u>**Note**</u>: Propulsion, chillers and thrusters will not be reconfigured after BO in harbour mode.

5.1.3.3. In manoeuvring mode

The restoration of the electrical distribution plant in manoeuvring mode will be achieved as follows:

After the first generator is connected on 11kV bus bar (Time: T1)) (refer to § 5.1.2.3)

<u>1st Step : supply of 440V switchboards</u>

Same as in harbour mode.

<u>2nd Step: supply of electric sub stations and main galley</u>

Same as in harbour mode.

<u>3rd Step: supply Propulsion</u>

When two generators are connected on 11kV bus bar (Time: T2)

Propulsion circuit breakers are controlled by Propulsion system and will be reclosed after black out when :

- Two (2) generators are connected on the bus bar of the corresponding propulsion circuit breaker (interlock done in propulsion control system)
- The operator put the propulsion levers references at 0 rpm
- The auxiliaries restart automatically by propulsion control cabinet 30 seconds after first generator connection in the HMS
- The operator reset the propulsion system in control panel

<u>Note</u>: Chillers and thrusters will not be reconfigured after BO in manoeuvring mode. Reconfiguration is performed by the operator.

5.1.3.4. In open sea mode

Same sequence as in manoeuvring mode.

5.1.4. Reconfiguration of ES1 and ES2

The reconfiguration of ES1 and ES2 is done by the operator when restarting after BO is completed.

5.1.4.1. Normal situation

<u>Initial situation : ES1 supplied by EG1 and ES2 supplied by EG2</u>.

<u>ES1 reconfiguration</u>:

These actions are performed BY OPERATOR:

1. Closing of Q01M11 if open (stay normally closed during BO)
2. Closing of QS1ES1 (synchronisation must be done)
   Opening after time delay of the EG1 circuit breaker QS2ES1 (local automation) Stop when cool down is achieved of EG1 (local automation)

The above sequence can be done either from EG1 room, synchronisation will be done by manual synchro or through MAS, synchronisation will be done by synchro unit located in ES1.

<u>ES2 reconfiguration</u>:

These actions are performed BY OPERATOR:

1. Closing of Q01M22 if open (stay normally closed during BO)
2. Closing of QS1ES2 (synchronisation must be done)
   Opening after time delay of the EG2 circuit breaker QS2ES2 (local automation) Stop when cool down is achieved of EG2 (local automation)

The above sequence can be done either from EG2 room, synchronisation will be done by manual synchro or through MAS, synchronisation will be done by synchro unit located in ES2.

5.1.4.2. Impaired situation

A) Initial situation : ES1 and ES2 supplied by EG2, ES1/ES2 bus ties are closed :

a) ES1/ES2 reconfiguration:

These actions are performed BY OPERATOR and have to be done when MS1 and MS2 are back power supplied :

1. Opening of ES1/ES2 bus ties that will lead black out of ES1

b) ES2 reconfiguration:

2. Closing command of CB QS1ES2 that will lead EG2 synchronising to MS2.
3. After load transfer of EG2, auto opening EG2 circuit breaker QS2ES2 and then auto stop EG2 after idling time. **All those actions are realised by local automation (except load transfer which is realised by MAS)**
4. Release of Trip commands for non-Solas consumers sent to ES2 (this when ES2 is power supplied by MS2).

c) ES1 reconfiguration:

5. Closing command of CB QS1ES1 that will lead ES1 alive and power supplied from MS1.
6. Release of Trip commands for non-Solas consumers sent to ES1 (this when ES1 is power supplied by MS1).

B) Initial situation : ES1 and ES2 supplied by EG1, ES1/ES2 bus ties are closed :

a) ES1/ES2 reconfiguration:

These actions are performed BY OPERATOR and have to be done when MS1 and MS2 are back power supplied :

1. Opening of ES1/ES2 bus ties that will lead black out of ES2

b) ES1 reconfiguration:

2. Closing command of CB QS1ES1 that will lead EG1 synchronising to MS1.
3. After load transfer of EG1, auto opening EG1 circuit breaker QS2ES1 and then auto stop EG1 after idling time. **All those actions are realised by local automation (except load transfer which is realised by MAS)**
4. Release of Trip commands for non-Solas consumers sent to ES1 (this when ES1 is power supplied by MS1).

c) ES2 reconfiguration:

5. Closing command of CB QS1ES2 that will lead ES2 alive and power supplied from MS2.
6. Release of Trip commands for non-Solas consumers sent to ES2 (this when ES2 is power supplied by MS2).

5.2. Half blackout

Half BO corresponds to the loss of one side (AFT or FWD) either for the 11kV or for the 440V.

5.2.1. Half blackout 11kV

5.2.1.1. FWD half blackout on HMS1

Initial situation :

- HMS1 and HMS2 are interconnected and fed by DG
- MS1 is supplied through FHTR1MS1 and FHTR2MS1
- MS2 is supplied through FHTR1MS2 and FHTR2MS2
- Emergency switchboard ES1 is fed via MS1 and ES2 is fed via MS2.

Half BO situation (loss of HMS1 which feed also ES1 through MS1):

This situation happens when over-current (short circuit) or electrical fault in one 11kV switchboard occurs. In this case the results will be the following :

1. Opening of all generator circuit breakers, opening of SWBD bus tie circuit breaker (CB locked OPEN. Local reset is required) which leads to HMS1 dead (and thus opening of all HMS1 outgoing circuit breakers by under voltage).
2. Opening of Q11HM1 (due to lack of voltage), which leads to MS1-1 dead and consequently ES1 dead.
3. Opening of Q12HM1 (due to lack of voltage), which leads to MS1-2 dead and consequently LMS1 dead.

Note : For auxiliaries with automatic change over which was running prior half blackout on power supply from MS1, the standby auxiliary power supplied from MS2 will be automatically started by the MAS.

Automatic power restoration of ES1, EMCC11-2, EMCC12-2 and substations loops :

1. Automatic starting and connection of EDG1 on ES1 (by local automation : blackout ES1 detection and ES1/ES2 busties not closed), that leads ES1 alive.
2. Bus tie EMCC11-1 / EMCC11-2 is opened by lake of voltage on EMCC11-1 (loss of normal power supply). When ES1 alive then EMCC11-2 will be back powered by ES1.
3. Bus tie EMCC12-1 / EMCC12-2 is opened by lake of voltage on EMCC12-1 (loss of normal power supply). When ES1 alive then EMCC12-2 will be back powered by ES1.
4. Automatic reconfiguration of the substation loops by the MAS, all substations which were connected on HMS1 prior to the half blackout will be connected, one by one, on HMS2.

Manual power restoration of MS1 :

- When cause of trouble on HMS1 is removed, then operator has to power supply MS1 through the FHTR1MS1 and FHTR2MS1 transformers.

Manual reconfiguration of ES1 :

The reconfiguration of ES1 will be possible when MS1 again becomes alive.

In this case, load transfer between EG1 and Normal 440V network (MS1) will be realised by the following :

- Closing command on CB QS1ES1 (*synchronisation to be done*) through MAS or locally on ES1.
- Automatic opening after time delay of EG1 circuit breaker QS2ES1 (local automation)
- Automatic stop of EG1 (local automation)

The above load transfer sequence can be done from EG1 room. Synchronisation will be done by manual synchro.

Synchronising through MAS can also be done (manual operation), synchronisation will be done by a synchro unit located in ES1.

5.2.1.2. Aft half blackout on HMS2

Initial situation :

- HMS1 and HMS2 are interconnected and fed by DG
- MS1 is supplied through FHTR1MS1 and FHTR2MS1
- MS2 is supplied through FHTR1MS2 and FHTR2MS2
- Emergency switchboard ES1 is fed via MS1 and ES2 is fed via MS2.

Half BO situation (loss of HMS2 which feed also ES2 through MS2):

This situation happens when over-current (short circuit) or electrical fault in one 11kV switchboard occurs. In this case the results will be the following :

1. Opening of all generator circuit breakers, opening of SWBD bus tie circuit breaker (CB locked OPEN. Local reset is required) which leads to HMS2 dead (and thus opening of all HMS2 outgoing circuit breakers by under voltage).
2. Opening of Q12HM2 (due to lack of voltage), which leads to MS2-2 dead and consequently ES2 dead.
3. Opening of Q11HM2 (due to lack of voltage), which leads to MS2-1 dead and consequently LMS2 dead.

Note : For auxiliaries with automatic change over which was running prior half blackout on power supply from MS2, the standby auxiliary power supplied from MS1 will be automatically started by the MAS.

Automatic power restoration of ES2, EMCC21-2, EMCC22-2 and substations loops :

1. Automatic starting and connection of EDG2 on ES2 (by local automation : blackout ES2 detection and ES1/ES2 busties not closed), that leads ES2 alive.
2. Bus tie EMCC21-1 / EMCC21-2 is opened by lake of voltage on EMCC21-1 (loss of normal power supply). When ES2 alive then EMCC21-2 will be back powered by ES2.
3. Bus tie EMCC22-1 / EMCC22-2 is opened by lake of voltage on EMCC22-1 (loss of normal power supply). When ES2 alive then EMCC22-2 will be back powered by ES2.
4. Automatic reconfiguration of the substation loops by the MAS, all substations which were connected on HMS2 prior to the half blackout will be connected, one by one, on HMS1.

Manual power restoration of MS2 :

- When cause of trouble on HMS2 is removed, then operator has to power supply MS2 through the FHTR1MS2 and FHTR2MS2 transformers.

Manual reconfiguration of ES2 :

The reconfiguration of ES2 will be possible when MS2 again becomes alive.

In this case, load transfer between EG2 and Normal 440V network (MS2) will be realised by the following :

- Closing command on CB QS1ES2 (*synchronisation to be done*) through MAS or locally on ES2.

- Automatic opening after time delay of EG2 circuit breaker QS2ES2 (local automation)
- Automatic stop of EG2 (local automation)

The above load transfer sequence can be done from EG2 room. Synchronisation will be done by manual synchro.

Synchronising through MAS can also be done (manual operation), synchronisation will be done by a synchro unit located in ES2.

5.2.2. Half blackout 440V

Initial situation:

- HMS1 and HMS2 are interconnected and fed by DG
- MS1 is supplied through FHTR1MS1 and FHTR2MS1
- MS2 is supplied through FHTR1MS2 and FHTR2MS2
- Emergency switchboard ES1 is fed via MS1 and ES2 is fed via MS2.

Two cases have to be taken into account for each FWD and Aft side. The first is a short circuit on MS1(MS2) and the second is the loss of MS1(MS2) power supply from HMS1(HMS2).

5.2.2.1. FWD half blackout 440V by loss of MS1-1

This situation can occur with over-current (short circuit) on MS1-1. The results are:

- Opening of secondary FHTR1MS1 which leads to MS1-1 dead and consequently ES1 dead.
- The MS1-1/MS1-2 bus tie must remain open (*).

Note : For auxiliaries with automatic change over which was running prior half blackout on power supply from MS1-1, the standby auxiliary power supplied from MS2 will be automatically started by the MAS.
(*) : Closing of MS1-1/MS1-2 bus tie is blocked if MS1-1 short circuit. MS1-1 short circuit corresponds to at least one of the two incoming CBs tripped (QS1M11 and MS1-1/MS1-2 bus tie).

Automatic power restoration of ES1 :

- Opening after time delay of incoming CB QS1ES1 in ES1 (local automation) which leads to automatic starting of EG1 and then automatic connection to ES1 (local automation).

Manual reconfiguration of ES1 :

The reconfiguration of ES1 will be possible when MS1-1 again becomes alive, same as description in § 5-2-1-1.

5.2.2.2. FWD half blackout 440V by loss of MS1-2

This situation can occur with over-current (short circuit) on MS1-2. The results are:

- Opening of secondary FHTR2MS1 which leads to MS1-2 dead and consequently LMS1 dead.
- The MS1-1/MS1-2 cross tie must remain open (*).

Note : For auxiliaries with automatic change over which was running prior half blackout on power supply from MS1-2, the standby auxiliary power supplied from MS2 will be automatically started by the MAS.

(*) : Closing of MS1-1/MS1-2 bus tie is blocked if MS1-2 short circuit. MS1-2 short circuit corresponds to at least one of the two incoming CBs tripped (QS1M12 and MS1-1/MS1-2 bus tie).

5.2.2.3. FWD half blackout 440V by loss of MS1-1 power supply

This situation occurs when the primary circuit breaker Q11HM1 of transformer FHTR1MS1 is tripped. The result is:

- Tripping of Q11HM1 will lead to MS1-1 dead and consequently ES1 dead.

Note : For auxiliaries with automatic change over which was running prior half blackout on power supply from MS1-1, the standby auxiliary power supplied from MS2 will be automatically started by the MAS.

Automatic power restoration of ES1 :

1. Opening after time delay of incoming CB QS1ES1 in ES1 (local automation) which leads to automatic starting of EG1 and then automatic connection to ES1 (local automation).
2. Bus tie EMCC11-1 / EMCC11-2 is opened by lake of voltage on EMCC11-1 (loss of normal power supply). When ES1 alive then EMCC11-2 will be back powered by ES1.

Manual power restoration of MS1 :

- Manual closing command by MAS of the MS1-1/MS1-2 bus ties circuit breaker (QBTMS1).

Manual reconfiguration of ES1 :

The reconfiguration Of ES1 will be possible when MS1-1 again becomes alive, same as description in § 5-2-1-1.

5.2.2.4. FWD half blackout 440V by loss of MS1-2 power supply

This situation occurs when the primary circuit breaker Q12HM1 of transformer FHTR2MS1 is tripped. The result is:

- Tripping of Q12HM1 will lead to MS1-2 dead and consequently LMS1 dead.

Note 1 : For auxiliaries with automatic change over which was running prior half blackout on power supply from MS1-2, the standby auxiliary power supplied from MS2 will be automatically started by the MAS.

Manual power restoration of MS1-2 :

- Manual closing command by MAS of the MS1-1/MS1-2 bus ties circuit breaker (QBTMS1).

5.2.2.5. Aft half blackout 440V by loss of MS2

Same description of MS1 from § 5-2-2-1 to § 5-2-2-4 has to be applied for MS2.

6. SPECIAL CONFIGURATION OF ES1 AND ES2

6.1. Black-out ES1 (ES2) with EG1 (EG2) blocked

In addition to cases already described in total black out and in half black out, there is one situation to be taken into account for power restoration of ES1 and ES2 when only one of both switchboards ES1 and ES2 was dead :

- Blackout ES1 (ES2) when EG1(EG2) is blocked

Status "EG1(EG2) blocked" is built with the following :

EG1(EG2) blocked = CB QS2ES1(QS2ES2) selector switch "OFF/ON" located on ES1(ES2) on position "OFF"

"OFF" corresponds to circuit breaker of EG1 out of order

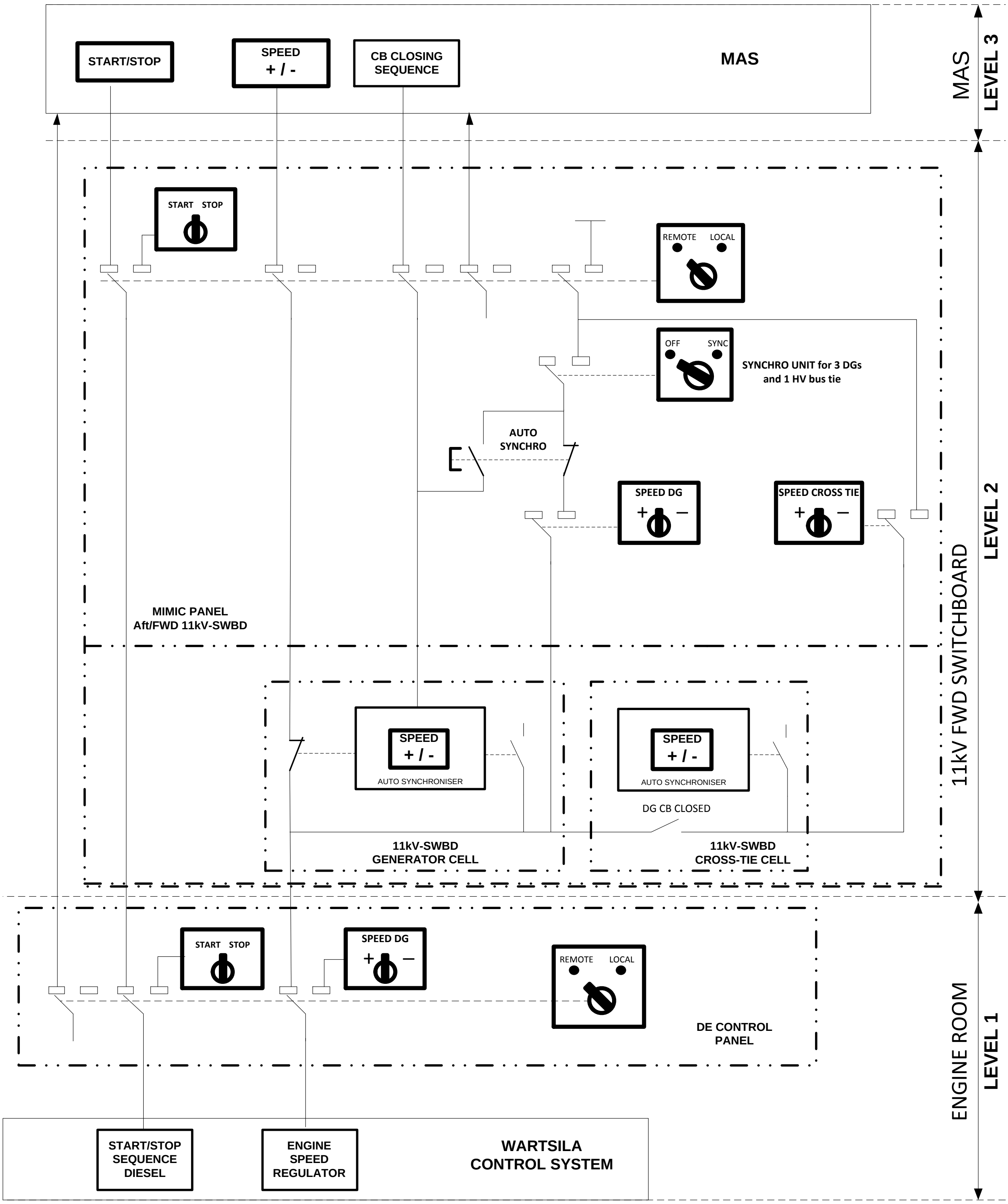
This situation can occur when for example half blackout on HMS1 (HMS2). The result is the following :

1. Trip commands for non-Solas consumers sent (MAS automatic control) to both emergency switchboards ES1 and ES2 ; this 3 sec after blackout detection on ES1 (ES2) bus bar, ES2(ES1) being alive.
2. Closing (MAS automatic control) of the ES1/ES2 bus ties 5 sec after blackout detection on ES1 (ES2) bus bar. Both emergency switchboards are now supplied by ES2 (ES1).

Note : Closing of ES1/ES2 bus ties is blocked if ES1(ES2) short circuit.

ES1(ES2) short circuit corresponds to at least one of the three incoming CBs tripped : QS1ES1, QS2ES1 and QCTES1 (QS1ES2, QS2ES2 and QCTES2).

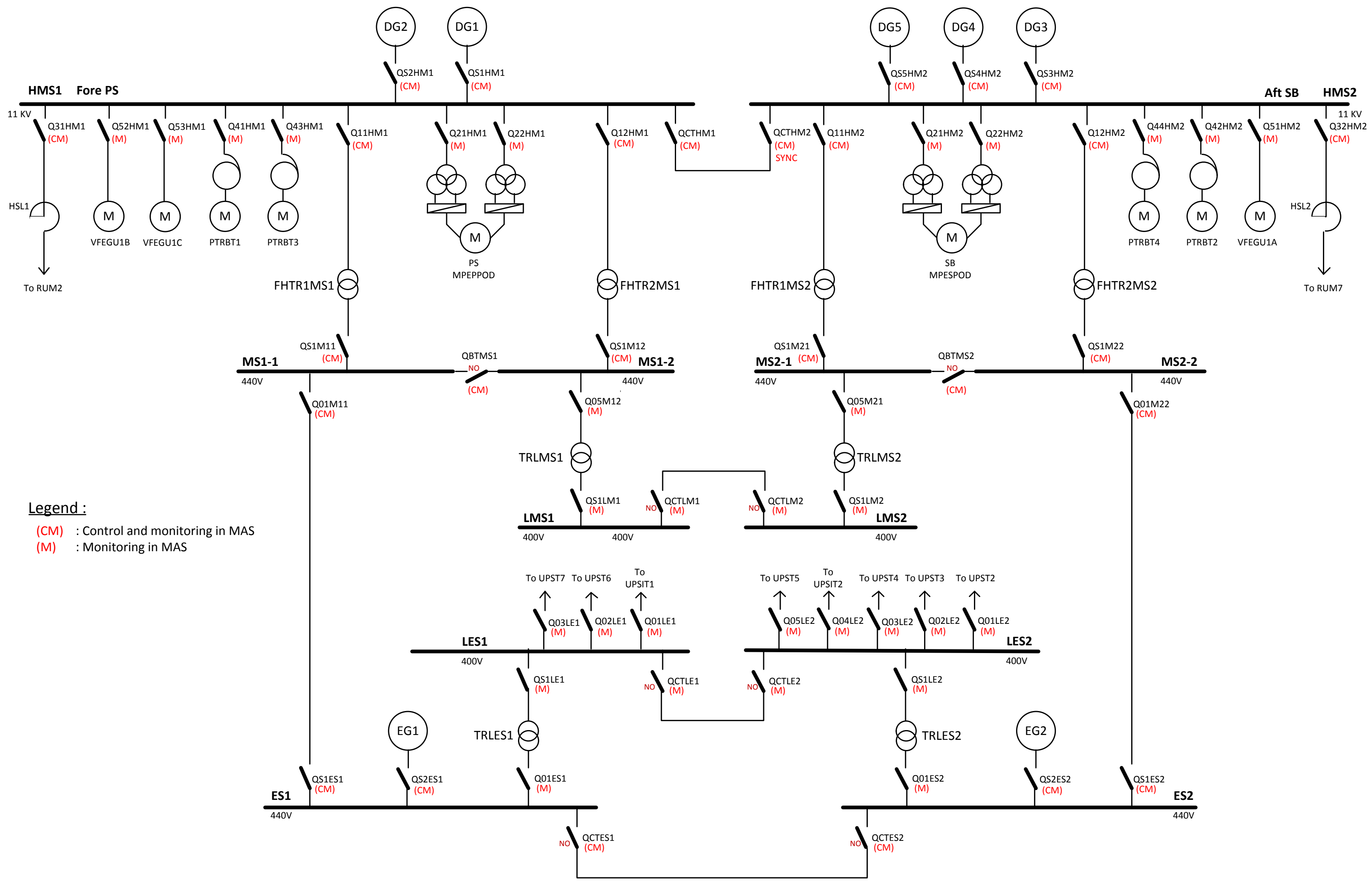
ANNEX 1 : DG - START/STOP and SPEED HIERARCHY



A34 start-stop-speed-hiera-DG

stx France	MAS – POWER PLANT PHILOSOPHY *MAS – PHILOSOPHIE DU SYSTEME DE PRODUCTION D'ENERGIE*	M40E4937	
		J34	Et suivants / *And next*
		Révision / *Revision* **A1**	Planche / *Sheet* **33**

ANNEX2 – ELECTRIC POWER PLANT OVERVIEW – P 1/2

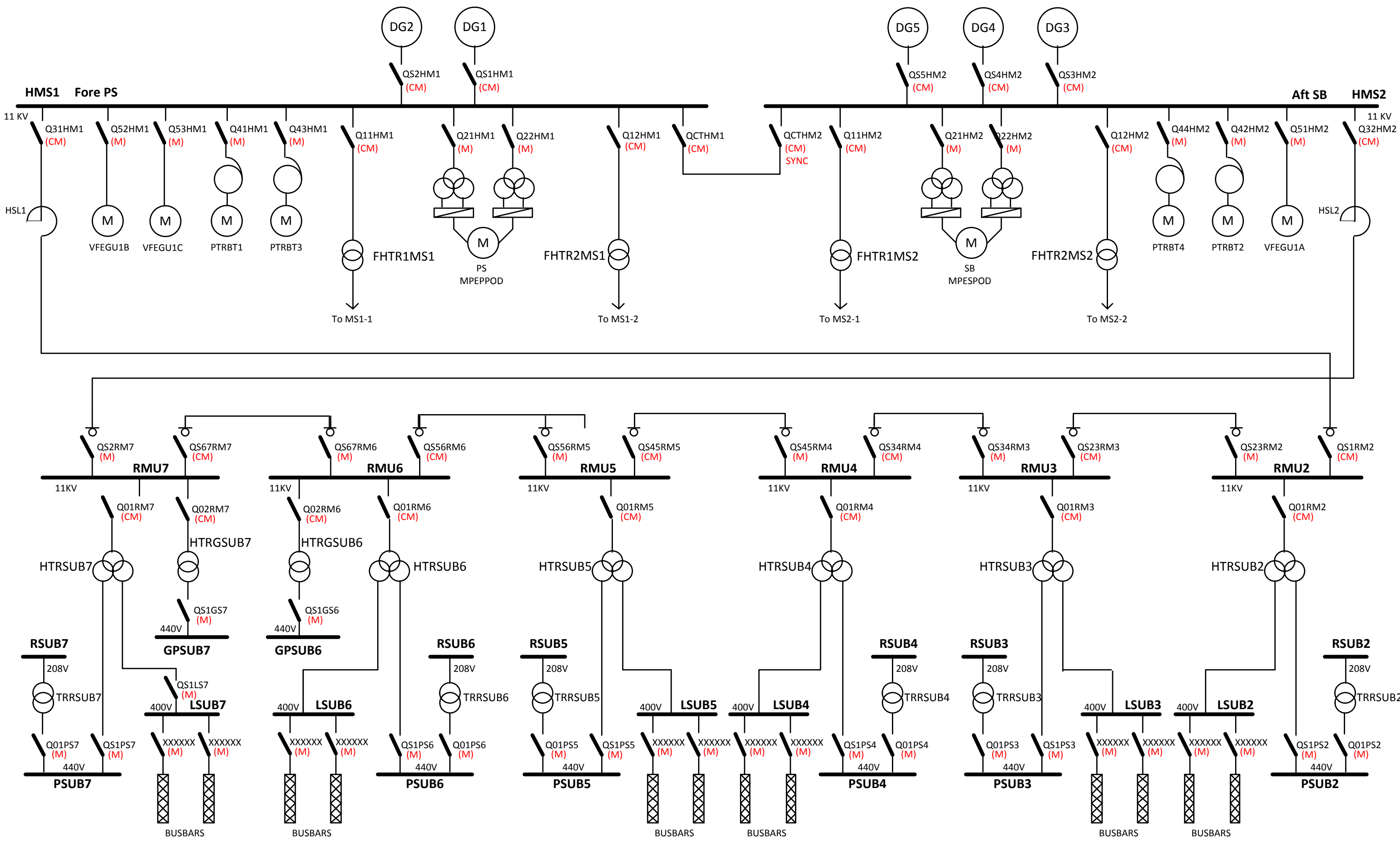


Legend :

(CM) : Control and monitoring in MAS
(M) : Monitoring in MAS

STX France	MAS – POWER PLANT PHILOSOPHY *MAS – PHILOSOPHIE DU SYSTEME DE PRODUCTION D'ENERGIE*	M40E4937	
		J34	Et suivants / *And next*
		Révision / *Revision* **A1**	Planche / *Sheet* **34**

ANNEX2 – ELECTRIC POWER PLANT OVERVIEW – P 2/2



Legend :

(CM) : Control and monitoring in MAS
(M) : Monitoring in MAS